

Intelligent Facility Automation & Industrial IoT Telemetry

A unified cloud-native platform providing sub-second SCADA visualization, predictive fault alarms, and complete 4M (Man, Machine, Material, Method) asset governance for commercial towers, residential townships, and industrial facilities.

< 500 msTELEMETRY
LATENCY**99.9%**SYSTEM
AVAILABILITY**35%**DOWNTIME
REDUCTION**100%**DIGITAL AUDIT
TRAIL

Executive Summary

Modern infrastructure facilities and smart buildings depend heavily on critical electromechanical systems—including booster water pumps, fire hydrants, sewage treatment plants (STP), and HVAC systems. Traditionally, these assets operate in silos, requiring manual paper logs and reactive repairs after catastrophic breakdowns occur. **Gmark IoT & 4M ERP** bridges this gap by marrying real-time edge telemetry (SCADA) with rigorous enterprise operational workflows, giving management, facility directors, and field engineers complete transparency over equipment health, energy draw, and operational safety.

The Core 4M Operational Framework

The platform is engineered around the proven industrial 4M methodology to guarantee holistic plant and facility management:

MAN**Workforce & Roles**

- Granular Role-Based Access (Super Admin to Operator)
- Direct Device & Facility User Assignment
- Shift logging & technician accountability
- Special permission overrides

MACHINE**Asset & Telemetry**

- 7-step lifecycle asset builder & specs tracking
- IoT edge controller linkage (MAC / Device ID)
- Real-time sensor feeds (bar, PSI, kW, Amps)
- Motor run hours & duty cycle rotation

MATERIAL**Spares & Inventory**

- Critical machine spare parts mapping
- Minimum inventory threshold alerts
- Replacement logs & supplier procurement data
- Consumable consumption tracking

METHOD**SOPs & Maintenance**

- Automated AMC & warranty tracking
- Preventative maintenance scheduling
- Digital document vault (manuals, circuit diagrams)
- Automated emergency trip SOP escalation

High-Level Architecture & Signal Flow

1. Field Sensors & PLC

Pressure transducers, flow meters, energy meters, AIPCU

**2. Edge IoT Gateway**

Modbus RS485 / Ethernet / 4G
GSM MQTT Publisher

**3. Gmark Core Engine**

NestJS Microservices,
PostgreSQL, Prisma, Event Bus

**4. Real-time SCADA UI**

React WebSocket Dashboard,
Multi-channel Alerts



Gmark IoT comes pre-packaged with industrial-grade, SVG-rendered SCADA templates specifically tuned for water, fire-fighting, and fluid automation systems. Each template offers high-frequency updates, interactive schematics, motor duty states, and live setpoint controls.

PUMPING

Domestic Terrace Booster System

Designed for residential and commercial multi-story pressure management. Features dynamic 3-pump or multi-stage booster monitoring, header discharge pressure gauge (0-10 bar), line trip protections, suction tank dry-run cutoff, and auto-duty alternation to balance motor wear and prevent coil burnout.

Auto / Manual Modes

VFD Frequency Hz

Header Pressure

SAFETY

Fire Fighting & Hydrant Panel

Mission-critical life-safety surveillance panel tracking Jockey Pump, Main Electric Hydrant Pump, and Diesel Standby Engine. Features real-time annunciator window alarms, battery voltage monitoring, engine cranking status, fuel tank level indicators, and NFPA/NBC compliant pressure threshold logging.

Diesel Cranking Status

Jockey Pump Cycling

Header Bar Alert

TRANSFER

Sump to Overhead Tank (OHT) Automation

Dual-reservoir automated liquid transfer control. Continuously computes sump suction levels and overhead tank availability using ultrasonic or magnetic level sensors. Automatically triggers transfer pump cycles, prevents dry-run cavitation, and signals overflow prevention interlocks.

High/Low Level Sensors

Overflow Protection

Runtime Counter

SUSTAINABILITY

Hydro-Pneumatic (HYPN) & Flushing Systems

Constant-pressure variable flow pumping for sanitary flushing and recycled greywater loops. Monitors energy consumption per cubic meter pumped, bladder tank pre-charge pressures, and detects instantaneous micro-leaks in building risers before internal water damage can occur.

Micro-Leak Detection

Energy (kWh) Tracking

Greywater Recycle

Telemetry Sensor & Parameter Matrix

System Module	Monitored Telemetry Parameters	Safety Interlocks	Control Options
Booster Pumps	Suction Pressure, Discharge Header Pressure, Motor Current (R/Y/B Amps), Voltage, VFD Speed	Dry Run Cutoff, Phase Failure, Overcurrent, High Pressure Burst	Auto / Manual / Remote Start-Stop / Setpoint Adjust
Fire Safety System	Main Hydrant Pressure, Jockey Pressure, Diesel Battery Volts, Fuel Level %, Engine Oil Temp	Low Pressure Alarm, Engine Fail-to-Start, Charger Fail	Emergency Alarm Silence, Test Mode, Manual Override
STP & Drainage	Sump High Level, Equalization Tank Level, Aeration Blower Run, Discharge Turbidity/TDS	Basement Flood Interlock, Thermal Overload Trip	Blower Timer Schedule, Auto Sludge Pump Cycle
Landscape Irrigation	Zone Solenoid Status, Line Flow Rate (LPM), Soil Moisture %, Weather Interlock	Pipe Burst Shutdown, No-Water Flow Lockout	Zone Scheduling, Remote Rain-Delay Mode

Instant Digital Twin Visualization

Every physical pump and pipe is mirrored in the UI using vector SVGs that illuminate dynamically based on live MQTT status: **Green (Running)**, **Grey (Standby)**, and **Flashing Red (Trip / Alarm)**. Users can diagnose equipment state at a single glance without visiting remote pump rooms or plant basements.



Multi-Tier Hierarchy & Enterprise Tenancy

The platform supports scalable multi-organization deployment, accommodating commercial corporate clients, residential real estate societies, and multi-location industrial conglomerates under one unified architecture:

1. Organization / Society

The top-level enterprise boundary. Defines corporate policies, global user pools, GST/billing compliance, and master administrative privileges.

2. Main & Sub-Facilities

Physical properties, corporate campuses, residential towers, or manufacturing units with localized facility heads and geospatial mapping.

3. Machines & IoT Nodes

Discrete equipment items linked to hardware telemetry controllers. Complete with specifications, AMC contracts, spare parts, and live telemetry feeds.

Automated Alerting & Incident Diagnostics

The platform monitors real-time telemetry against user-configured threshold rules and escalates breaches instantly through multiple channels:

Automated Multi-Channel Notifications

Configurable alert triggers evaluate parameter thresholds (e.g. pressure below 2.0 bar, motor trip, thermal breach). Instant HTML email summaries with device location, current values, and direct SCADA dashboard deep-links are dispatched to assigned engineers and technicians.

Fault History & Snapshot Logging

Every alarm incident automatically captures a sub-second time-series snapshot of all registers before and after the event. This forensic log allows maintenance teams to pinpoint root causes (e.g., power surge vs. pipe burst) without guesswork.

Role-Based Access Control (RBAC) & Special Permissions

Security is enforced at every layer. Users only see and control the facilities and machines explicitly delegated to their profile:

Super Admin

Full global visibility across all societies, main facilities, user credentials, system configs, and master permission templates.

Facility Admin

Supervises all towers, pump rooms, employee shift assignments, and operational maintenance for their assigned complex.

Maintenance Eng.

Manages asset lifecycle, spare part inventories, AMC contract renewals, and responds to high-priority equipment alerts.

Device Operator

Dedicated access strictly locked to specific machines (e.g., booster pump operator). Prevents accidental cross-facility errors.

Enterprise Security & Compliance Standards

Built with state-of-the-art web security: **JWT authentication with rotating access tokens**, bcrypt password hashing, encrypted TLS WebSocket pipelines, and automated tamper-evident Audit Logging tracking every user click, export, and configuration edit.



Measurable Business Value & ROI

Up to 35% Downtime Cut

Predictive alerts warn of abnormal pressure cycling and phase imbalances before pump stator burnout or burst pipes occur.

20-30% Energy Savings

Automated duty cycling and VFD setpoint optimization prevent unnecessary continuous motor operation during off-peak hours.

100% Paperless Audits

Eliminates error-prone physical clipboards. Historical telemetry, maintenance logs, and AMC contracts are instantly exportable to PDF and Excel.

Platform Technical Specifications

Architectural Layer	Technical Specification & Capabilities
Edge Protocols	MQTT (v3.1.1 / v5.0), Modbus RTU over RS-485, Modbus TCP/IP, AIPCU Controller protocol, HTTP REST Webhooks
Supported Hardware	Siemens, Schneider Electric, ABB, Delta, Custom IoT Microcontroller Gateways, Ultrasonic/Hydrostatic Level Transmitters
Cloud Core Engine	NestJS (Enterprise TypeScript), Prisma ORM, PostgreSQL with timescale indexing, Redis pub/sub queueing
Frontend SCADA	React 18+, TypeScript, Tailwind CSS, Native Canvas/SVG Telemetry Visualizers, Socket.io Real-Time Client
Deployment Modes	Multi-tenant Cloud SaaS (AWS / Azure / GCP), Private Cloud Dedicated Cluster, or Local On-Premise Server deployment
Data Export & APIs	Automated CSV / Excel / PDF report generation, RESTful APIs for ERP / SAP integration, Third-Party BI connectors

Key Target Industries & Implementations

Smart Real Estate

High-rise residential societies, commercial IT parks, gated communities.

Industrial Plants

Textile mills, automotive lines, food & beverage processing, utility rooms.

Healthcare & Hotels

Continuous water supply, hot water recirc loops, fire compliance monitoring.

Municipal Utilities

Sewage treatment plants (STP), water pumping stations, storm-water sumps.

Ready to Transform Your Facility Operations?

Schedule a live proof-of-concept demonstration or connect with our IoT solution architects.

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